CBSE MOCK — Pre-Board Practice Series

Central Board of Secondary Education

CLASS XII — BIOLOGY

Date: 12 November 2026 | Maximum Marks: 70 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. This question paper contains 5 sections. Read the instructions for each section carefully.
2. All questions are compulsory unless stated otherwise.
3. Write all answers neatly and show relevant working wherever required.
4. Marks for each question are indicated against it.
5. This paper is a Practice Paper — it is similar to but NOT an actual CBSE board paper.
6. Download more FREE practice papers daily at: **www.cbsemock.org.in**

SECTION A — MCQ & Assertion-Reason (16 Marks)

MCQ (1 mark each) | A/R (1 mark each)

Q1. **[1 mark]**

Which of the following is used as a vector in recombinant DNA technology for plant cells?

(A) Agrobacterium tumefaciens
(B) E. coli
(C) Rhizobium
(D) Yersinia

Q2. **[1 mark]**

Which antibiotic is produced by Penicillium notatum?

(A) Streptomycin
(B) Penicillin
(C) Tetracycline
(D) Chloramphenicol

Q3. **[1 mark]**

The 'cry' genes found in Bt cotton produce proteins that are toxic to:

(A) fungi
(B) bacteria
(C) insects
(D) viruses

Q4. **[1 mark]**

Species found in a small geographic area and nowhere else are called:

(A) keystone species
(B) invasive species
(C) endemic species
(D) indicator species

Q5. **[1 mark]**

Which human chromosome is acrocentric?

(A) Chromosome 1
(B) Chromosome 13
(C) Chromosome 3
(D) Chromosome X

Q6. **[1 mark]**

Assertion (A): In adaptive radiation, several different species evolve from a single ancestral taxon in a short time. Reason (R): Different species radiate into new environments and evolve adaptations to exploit new ecological niches.

Q7. **[1 mark]**

The 'primary productivity' of an ecosystem depends on:

(A) rate of photosynthesis only
(B) rate of photosynthesis minus rate of respiration
(C) rate of assimilation of organic matter
(D) biomass at a given time

Q8. **[1 mark]**

Restriction enzymes are obtained from:

(A) eukaryotes
(B) prokaryotes
(C) viruses
(D) fungi

Q9. **[1 mark]**

Which of the following is an in situ conservation method?

(A) Botanical garden
(B) Wildlife sanctuary
(C) Seed bank
(D) Zoo

Q10. **[1 mark]**

In a food chain, only about what percentage of energy is transferred from one trophic level to the next?

(A) 1%
(B) 5%
(C) 10%
(D) 20%

Q11. **[1 mark]**

In Meselson-Stahl experiment, E. coli cells were grown in ^{15}N medium and then transferred to ^{14}N medium. After two generations, the DNA would show:

(A) all heavy
(B) all light
(C) half heavy-light, half light
(D) all heavy-light

Q12. **[1 mark]**

The number of nucleotides present in a typical tRNA molecule is approximately:

(A) 20
(B) 60
(C) 80
(D) 120

Q13. **[1 mark]**

Assertion (A): RNA can act as a genetic material as well as a catalyst. Reason (R): RNA has a 2'-OH group that makes it less stable than DNA.

Q14. **[1 mark]**

Which of the following is a correct match?

(A) ABO blood group — single gene with two alleles
(B) Sickle cell anaemia — chromosomal disorder
(C) Down syndrome — trisomy of chromosome 21
(D) Haemophilia — autosomal dominant

Q15. **[1 mark]**

In which phase of meiosis do homologous chromosomes undergo synapsis?

(A) Leptotene
(B) Zygotene
(C) Pachytene
(D) Diplotene

Q16. **[1 mark]**

Hardy-Weinberg equilibrium is disturbed by:

(A) random mating
(B) large population size
(C) genetic drift
(D) no mutation

SECTION B — Very Short Answer (10 Marks)

VSA (2 marks each)

Q17. **[2 marks]**

Distinguish between primary and secondary succession.

Q18. **[2 marks]**

Draw a labelled diagram of a mature embryo sac of a flowering plant and name the cells that form the endosperm after fertilization.

Q19. **[2 marks]**

What is 'gene therapy'? Name the first clinical application of gene therapy in humans. OR Explain the principle of PCR. Name the thermostable DNA polymerase used.

Q20. **[2 marks]**

(a) What is 'lac operon'? Name its structural genes. (b) Which molecule acts as an inducer in the lac operon? OR What is the difference between euchromatin and heterochromatin? Which form is transcriptionally active?

Q21. **[2 marks]**

Name the type of immunity involved when a mother passes antibodies to her foetus through the placenta. Write one difference between active and passive immunity.

SECTION C — Short Answer (21 Marks)

SA (3 marks each)

Q22. **[3 marks]**

Explain with the help of a diagram the construction of a recombinant DNA molecule using the restriction enzyme EcoRI. Name the type of ends formed.

Q23. **[3 marks]**

Explain with examples: (a) Competitive exclusion principle (b) Resource partitioning (c) Keystone species

Q24. **[3 marks]**

Describe the causes and effects of: (a) Eutrophication (b) Biomagnification (c) Ozone depletion

Q25. **[3 marks]**

Draw a labelled diagram showing the life cycle of a Plasmodium. Name the two hosts. What symptoms are associated with the release of merozoites? OR What is Acquired Immunodeficiency Syndrome (AIDS)? How is it transmitted? Name the drug used in its treatment.

Q26. **[3 marks]**

In a monohybrid cross between two pure-breeding individuals for tallness (TT) and dwarfism (tt): (a) Find the phenotypic ratio in F_2 (b) Find the genotypic ratio in F_2 (c) What is the probability of getting a homozygous dominant individual in F_2?

Q27. **[3 marks]**

Explain the process of biogas production. Name the organisms involved and write the overall chemical reaction.

Q28. **[3 marks]**

Describe the following with examples: (a) Allogamy (b) Autogamy (c) Geitonogamy Which of these promotes outbreeding?

SECTION D — Case Study (8 Marks)

CBQ (4 marks each — Case Study)

Q29. **[4 marks]**

Biodiversity refers to the variety of life at all levels from genes to ecosystems. India is a mega-diversity nation with about 8% of global biodiversity. The IUCN Red List categorizes species based on extinction risk. Currently, about 1 million plant and animal species face extinction. Major causes include habitat loss (HIPPO: Habitat loss, Invasive species, Pollution, Population growth, Overexploitation). (a) What are the different levels of biodiversity? (b) State the importance of biodiversity conservation. (c) Distinguish between in situ and ex situ conservation with examples. OR (c) Explain the 'Rivet Popper' hypothesis related to biodiversity loss.

Q30. **[4 marks]**

The Human Genome Project (HGP) was a massive international scientific effort completed in 2003. It sequenced all 3.2 billion base pairs in the human genome. HGP used ESTs (Expressed Sequence Tags) for gene discovery and created a detailed genetic and physical map. It found approximately 30,000 genes. The sequence is 99.9% identical among humans. (a) What are ESTs and how were they used in HGP? (b) State any two salient features of the human genome. (c) What is meant by bioinformatics? Name one database used to store genomic sequences. OR (c) How did HGP contribute to the development of personalized medicine?

SECTION E — Long Answer (15 Marks)

LA (5 marks each)

Q31. **[5 marks]**

(a) Describe the process of fertilization in a flowering plant. What are double fertilization and triple fusion? (b) Draw a labelled diagram of a mature seed of Phaseolus (bean). (c) Explain parthenocarpy and apomixis. OR (a) Describe the events that occur in the female reproductive system from ovulation to implantation. (b) Explain in vitro fertilization (IVF). What is GIFT? (c) State any three advantages of lactational amenorrhoea as a contraceptive method.

Q32. **[5 marks]**

(a) Explain the central dogma of molecular biology. (b) Describe the process of translation in eukaryotes with the help of a diagram. (c) What is the significance of the wobble hypothesis? OR (a) Describe the structure of DNA as proposed by Watson and Crick. Include the base pairing rules. (b) What is the significance of the antiparallel nature of the two DNA strands? (c) Explain the semi-conservative nature of DNA replication with Meselson-Stahl experiment.

Q33. **[5 marks]**

(a) What are transgenic animals? Give two examples and their applications. (b) Explain the following: (i) Bt toxin mechanism of action (ii) RNAi (RNA interference) as pest control (c) What are biosafety concerns related to GMOs? Name the regulatory body in India. OR (a) What is somatic hybridization? Explain with the example of Pomato. (b) Explain recombinant DNA technology step by step, from isolation of gene to expression. (c) What are molecular markers? How are they useful in crop improvement?

✦ END OF QUESTION PAPER ✦

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